

Lumen

The complete AI-powered research workflow.

Stop cobbling together search, summarization, extraction, and analysis tools. Lumen handles search, summarization, extraction, and analysis in one place.

Ask: How do RAG systems improve medical QA?

Search Corpus

The Academic Bottleneck

The current state of literature review is broken.

2,000_{hrs}

Systematic Review

Researchers lose weeks searching across sources, manually building comparison tables, and worrying about missing critical work.

72%

The LLM Problem

Existing AI tools hallucinate citations at a 30-72% failure rate. Academic integrity is compromised, and you simply cannot trust the final product.

The Failure Chain

1. Can't find papers

2. Can't synthesize cross-paper

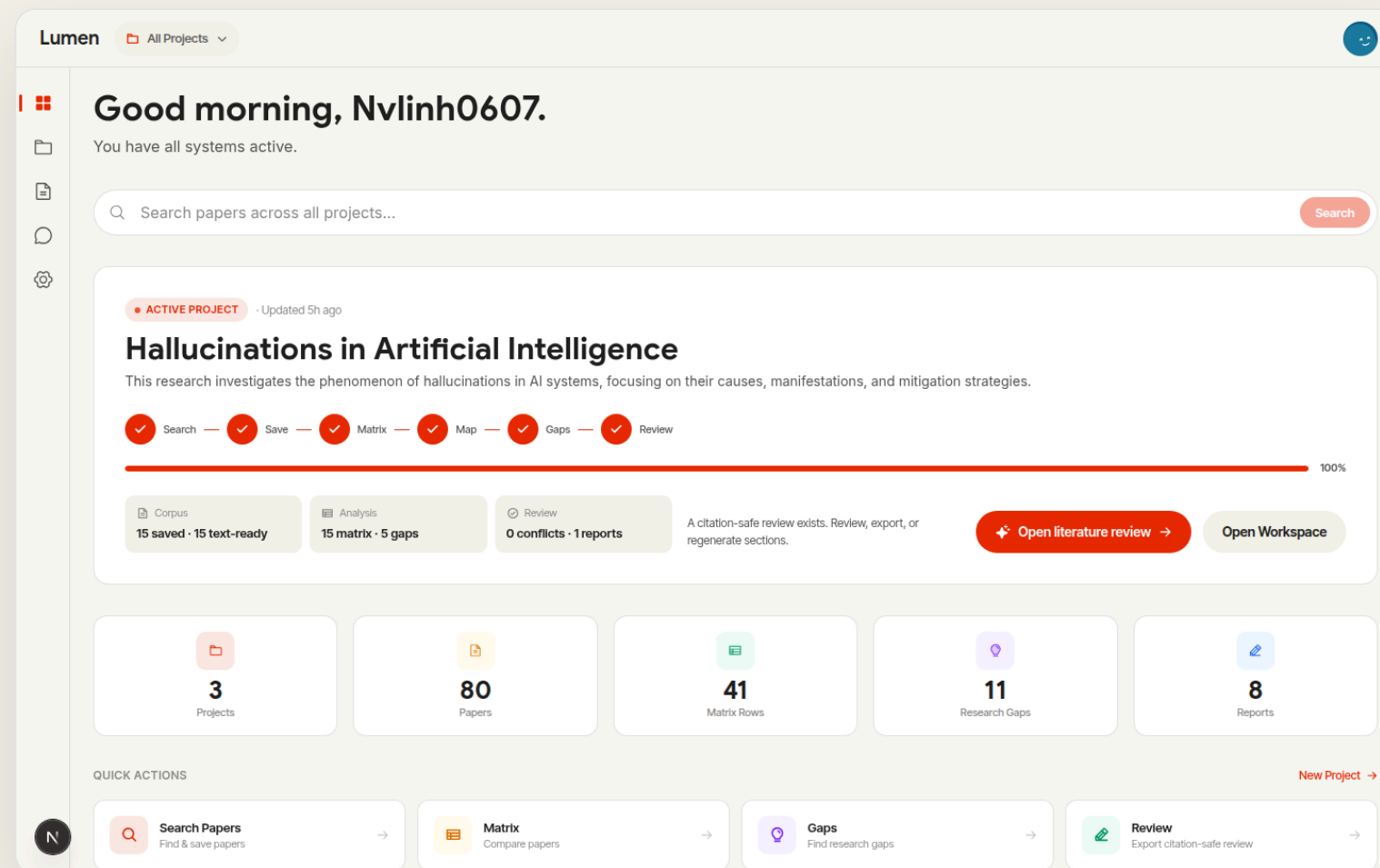
3. AI invents citations

4. Cannot verify evidence

5. Trust broken.

Your Research Hub

A clean, focused environment for discovery and synthesis.



SINGLE PANE OF GLASS

All projects, papers, and progress in one view

CROSS-PROJECT SEARCH

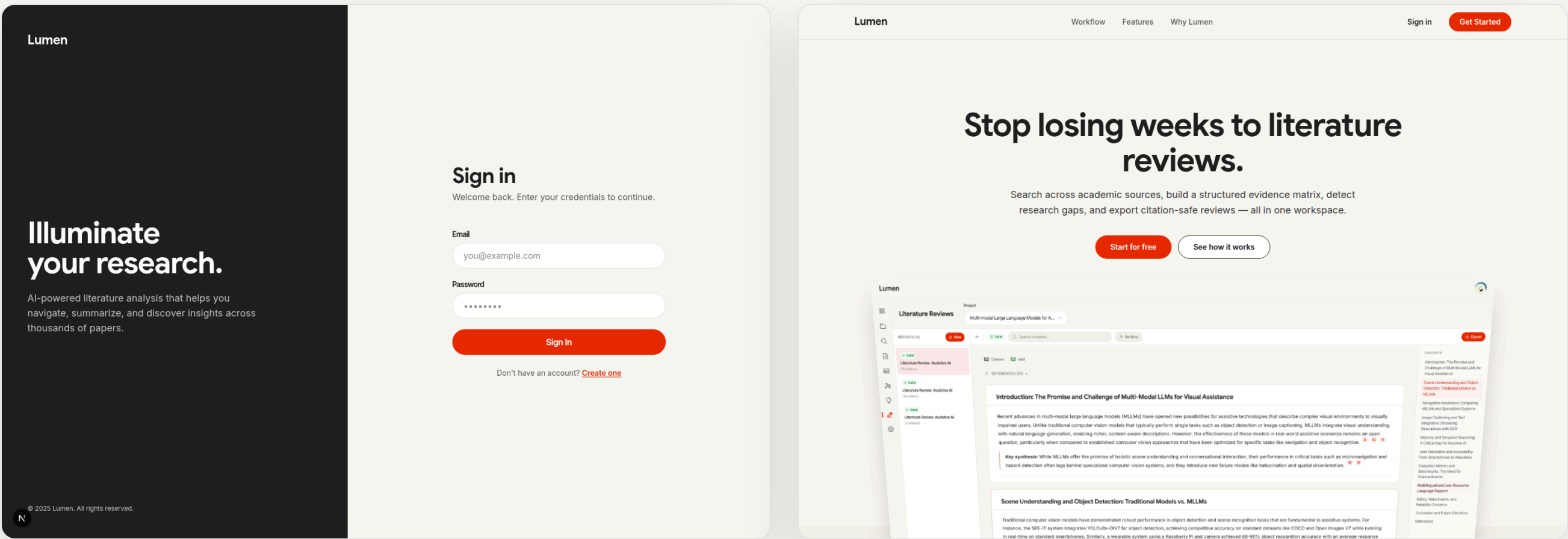
Find papers across your entire library instantly

REAL-TIME PROGRESS

Track every pipeline stage as it runs

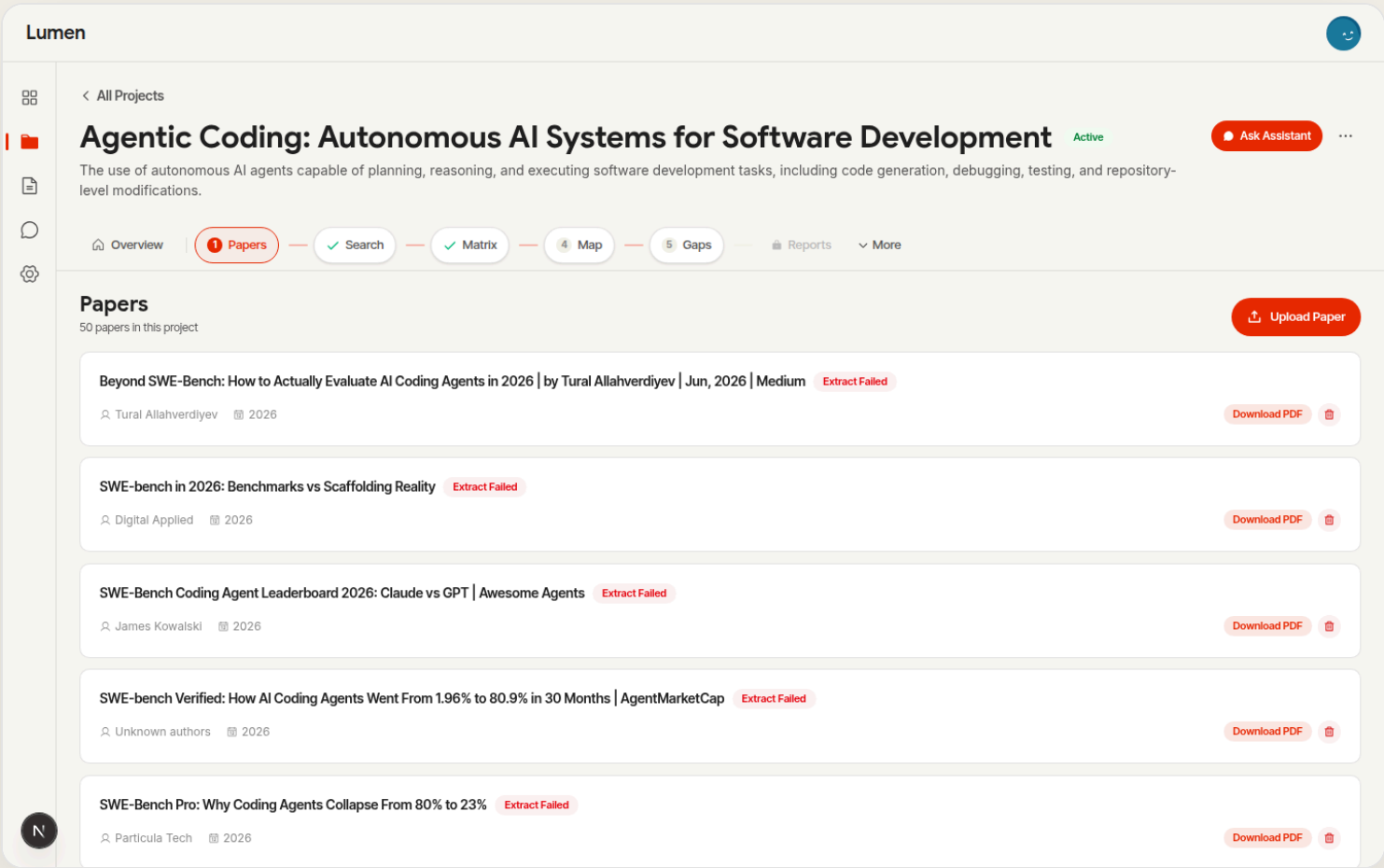
Seamless Onboarding

From secure login to the main workspace in seconds.



Multi-Source Paper Search

Pull from Semantic Scholar, arXiv, OpenAlex, Exa, and Firecrawl—all deduplicated.



5

5 academic sources, 1 corpus

Every paper is automatically deduplicated by DOI, arXiv ID, and Semantic Scholar ID. Save 10-15 highly relevant papers in seconds.

AUTO-DEDUPLICATION

Same paper from multiple sources =
1 entry

The Literature Matrix

Auto-extracted, editable, transparent.

7 FIELDS EXTRACTED

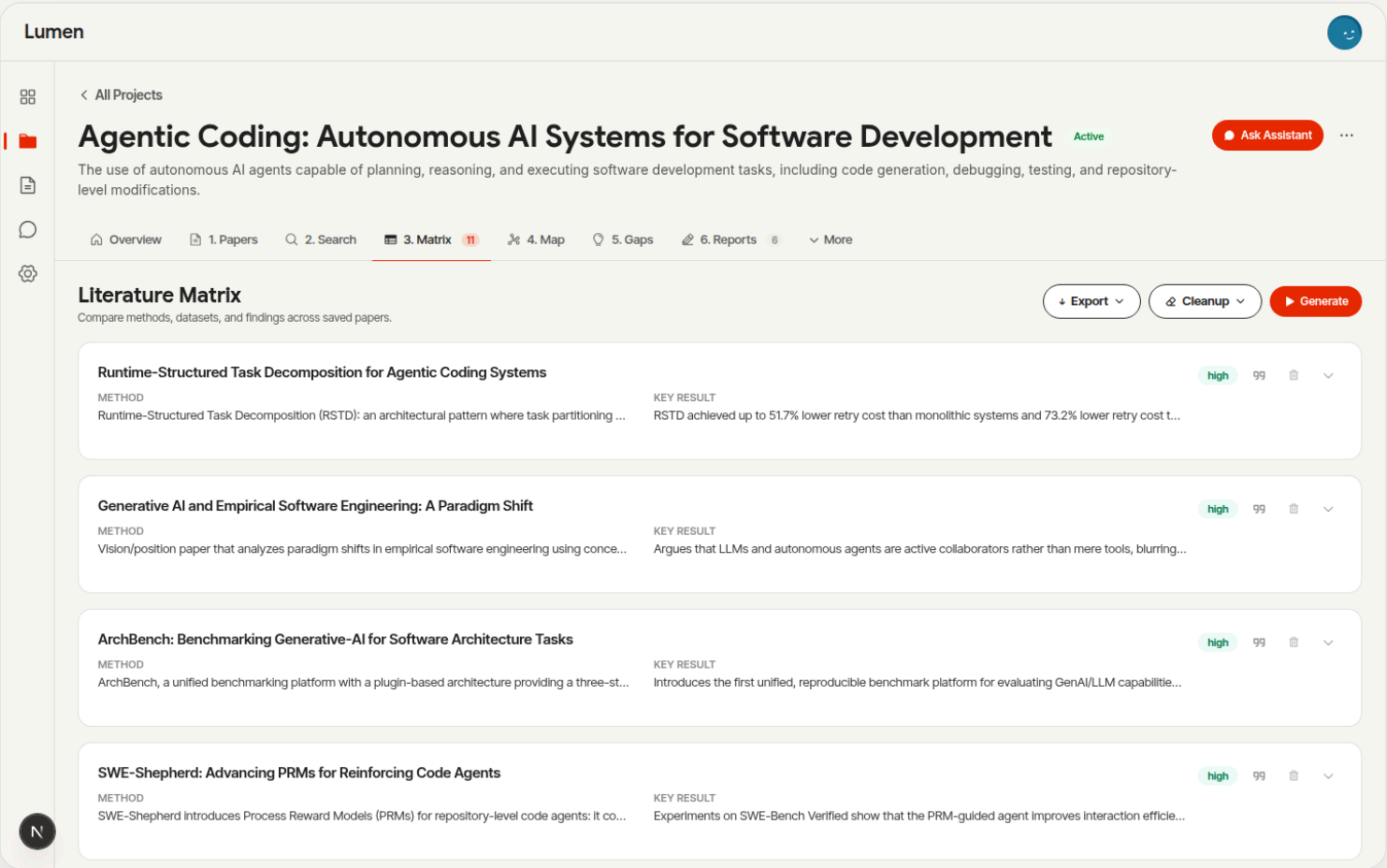
Method • Dataset • Result •
Limitation • ...

CONFIDENCE-TAGGED

Every row shows AI certainty

FULLY EDITABLE

AI gets it wrong? Click to fix in 1
second



8 Core Capabilities

The complete research lifecycle wrapped in a single application.

1

Auth & Projects

Secure JWT user management and private project containers for isolated research.

2

Multi-Source Search

Live integration with Semantic Scholar, OpenAlex, arXiv, Exa, and Firecrawl.

3

Save & Screen

AI-assisted relevance scoring immediately highlights high-value papers for your corpus.

4

Auto-Matrix

Fully editable literature matrix with automated LLM extraction of methodology and findings.

5

Knowledge Map

Interactive force-directed graph visualization of connected concepts and datasets.

6

Gap Detection

Automatically identify underexplored academic areas strictly backed by hard evidence.

7

Citation Guardrail

A strict backend-validation layer automatically blocks hallucinated reference IDs.

8

Markdown Export

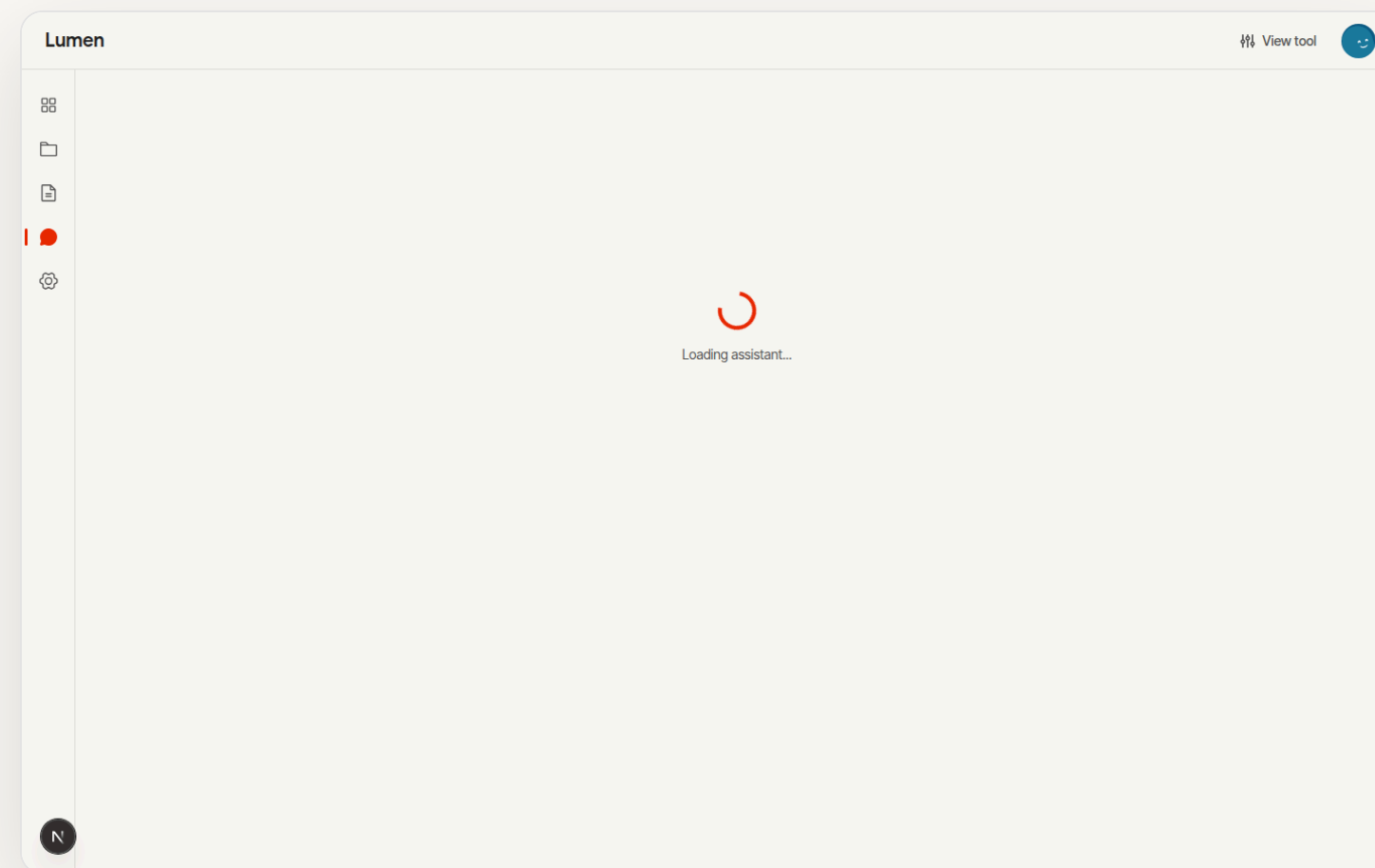
Generate ready-to-publish draft literature reviews with pristine, accurate citation trails.

The Lumen Chat Assistant

Drive the entire workflow through natural conversation.

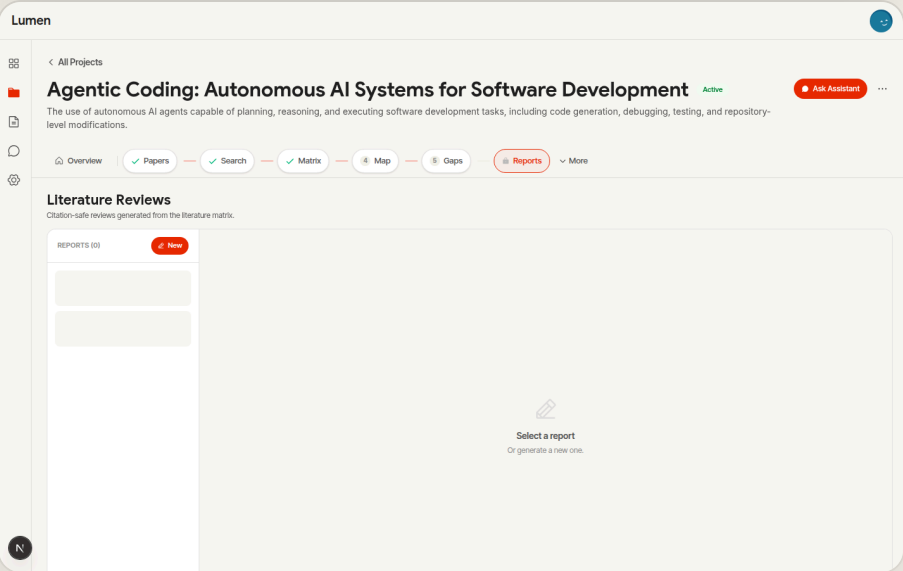
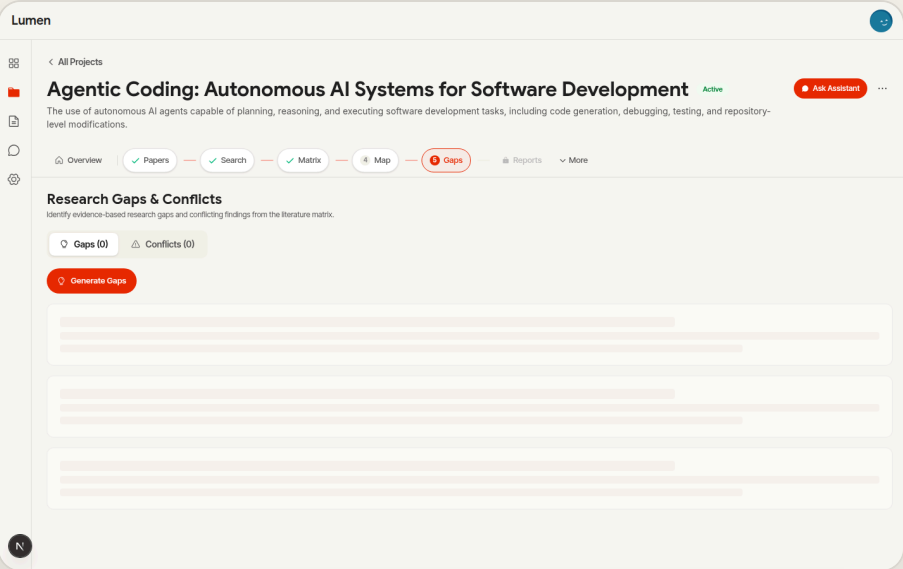
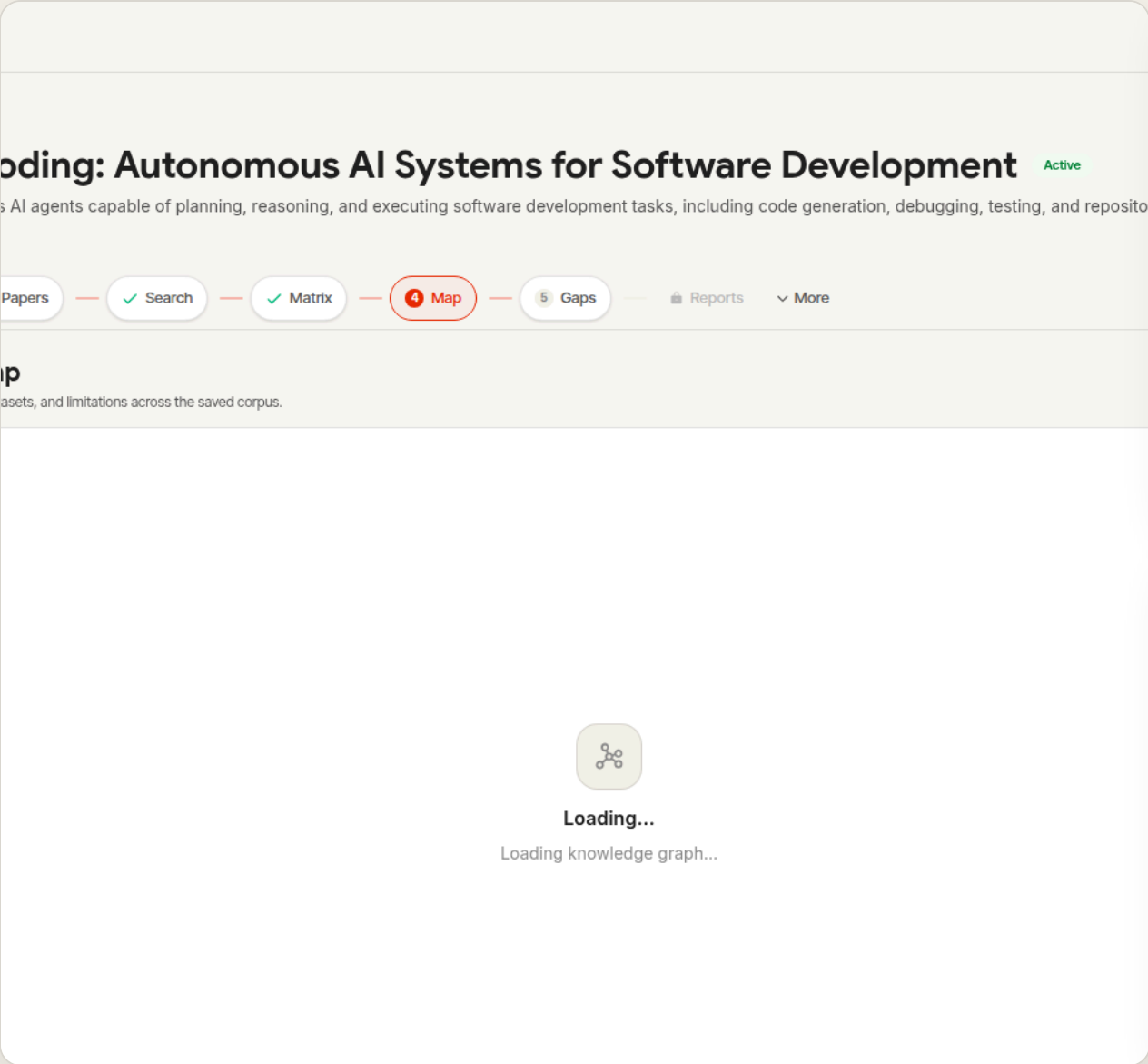
Instead of clicking through 7 separate pages, command the Assistant to execute the end-to-end pipeline in one interface.

- ✓ **Plan-Act Transparency:** Watch the agent plan and execute live.
- ✓ **Zero Data Duplication:** Artifacts save directly to your workspace.
- ✓ **Full Guardrails Intact:** Strict validation blocks hallucinations.



Map & Gap Detection

Visualize connections. Discover what's missing.



0% Hallucinations.

The 3-Layered Defense System.

Our proprietary citation guardrail enforces absolute data purity. If a claim isn't in your curated project corpus, Lunien refuses to cite it.

1. Preventive

UI blocks and actively warns the user when attempting to save papers with "low" AI-relevance scores to the project corpus.

2. Detective

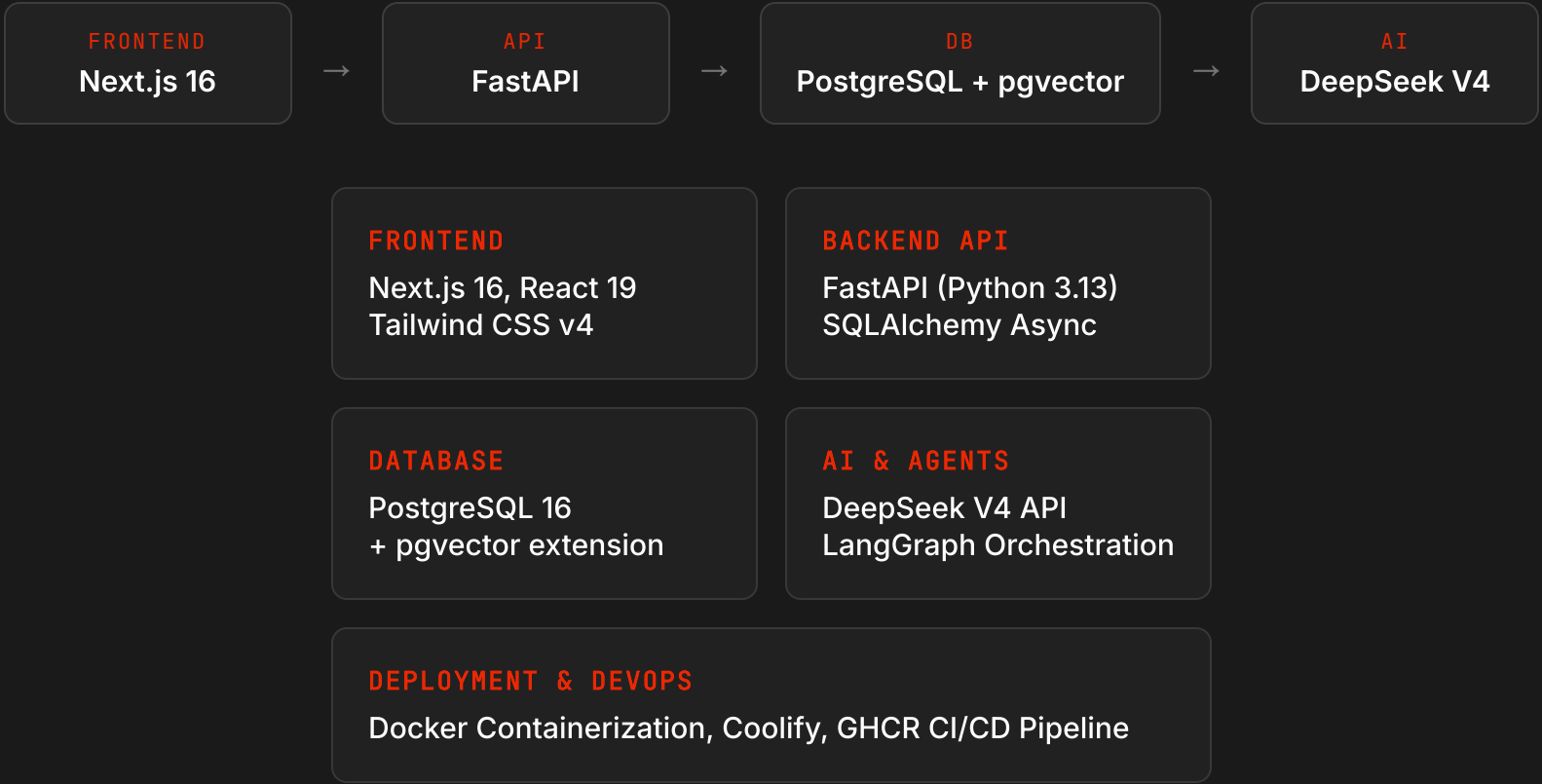
Matrix auto-tagging visually highlights rows where AI extraction confidence is low, enabling 1-click removal of bad data.

3. Protective

Backend Hybrid RAG explicitly drops DB records flagged as irrelevant. Bad data never reaches the final LLM report step.

Modern Architecture

Built for scale, speed, and intelligent orchestration.



Unit Economics

High performance. Insanely low cost.

By leveraging highly optimized LLMs (DeepSeek V4, Claude 3 Haiku) and localized embeddings, Lumen remains highly accessible to academic institutions.

80⁺

Papers Indexed

41

Matrix Rows Extracted

11

Research Gaps Detected

8

Citation-Safe Reports

0%

Hallucinated Citations

Ready to redefine synthesis?

Deploy your private research assistant today.

[View GitHub](#)[Read Documentation](#)

Thank You.

Questions & Answers

CONTACT

hello@lumen.com

SOURCE

github.com/lumen-ai